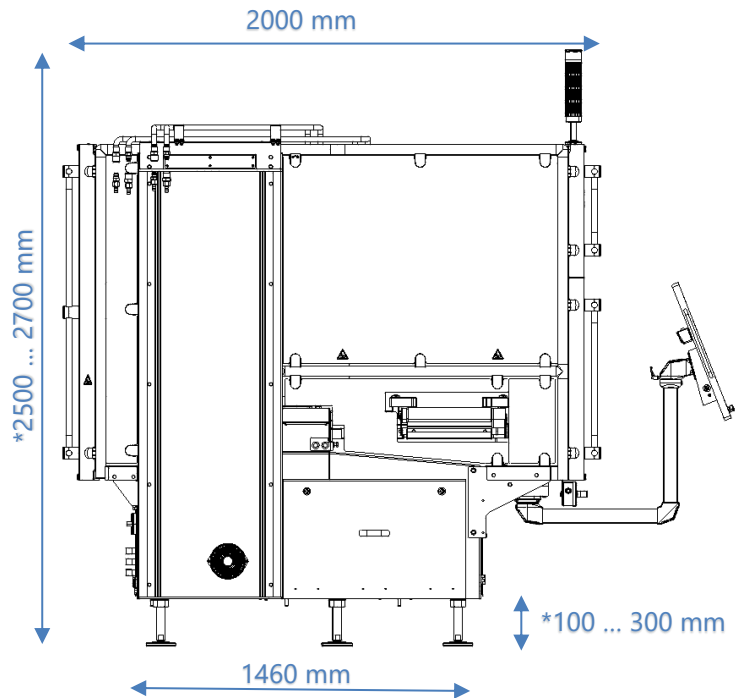
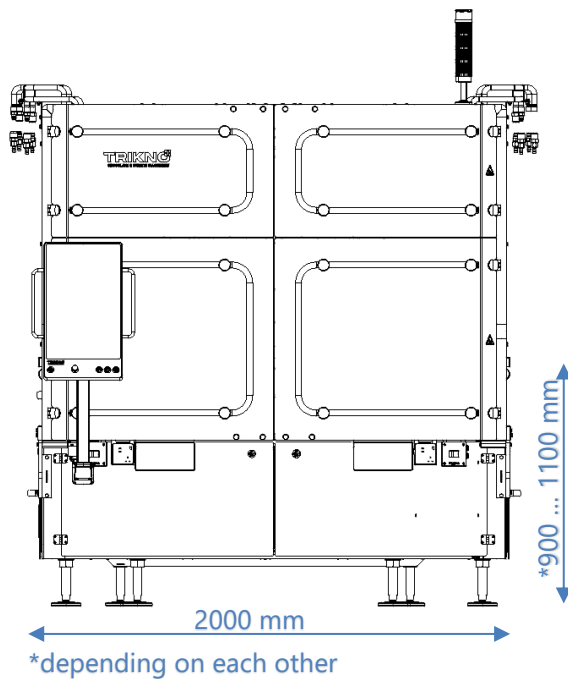
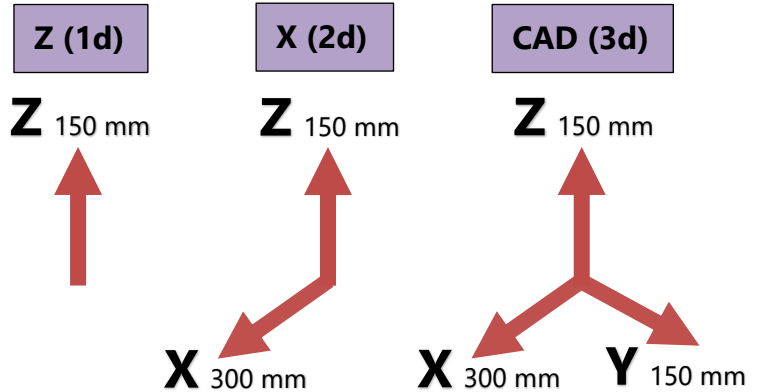
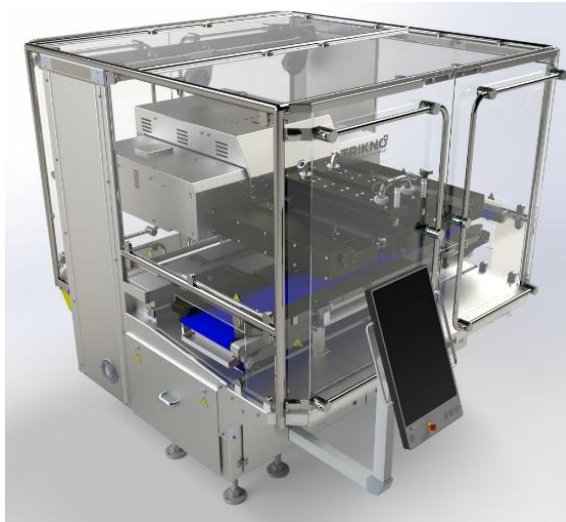




Technical Data Sheet TCM Alpha

Machine with proven single-rotary piston system, fully powered by electric servo motors.
Wide range of applications.



Single	9, 12	16, 18, 24	32, 36, 48	Optional
Boxer	9/18, 12/24	16/32, 18/36, 24/48	32/64, 36/72, 48/96	For detailed information, see variants further down in the document
Liner width	35 mm	45 mm	30 mm	
Piston distance	35 mm	22.5/45 mm	11 mm	
Piston stroke	52 mm	60 mm		
Conveyor length	2'000 mm			... 3'750 mm
Mould widths	261 ... 412 mm			... 517 mm
Weight	2,000 kg	2,250 kg	2,500 kg	



Features

- ✓ Head movement by electric servo motors possible as single-axis (vertical: Z), two-axis (vertical and belt direction: X) or three-axis (CAD)
- ✓ High-precision single rotary piston system with identical dosing volumes (proven for more than 30 years – Single Piston System)
- ✓ No mixing between the pistons possible (1 piston = 1 nozzle)
- ✓ Machine designed for industrial application
- ✓ Quick changeover of different depositing tools (pistons/liners/nozzle plates) without additional tools
- ✓ Number of pistons individually selectable (easy activation / deactivation by using individual couplings)
- ✓ Easy cleaning of the dosing machine without additional tools (Stainless steel or aluminium in ALTEF® coating – for pH values 4 ... 8)
- ✓ Easy cleaning of the pistons and cylinder liners (no tools necessary)
- ✓ Movable touchscreen HMI for machine operation
- ✓ Depositing head in free-arm design for easy operation / cleaning / adjustment
- ✓ Fully digital machine control
- ✓ Remote access on PLC, HMI, servo drives possible
- ✓ Software designed for maximum accuracy and flexibility
- ✓ One Shot software already installed on every machine
- ✓ Double-walled hopper for the chocolate side

Options

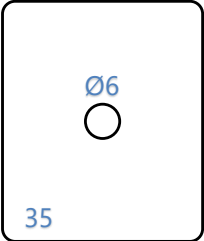
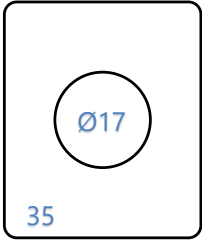
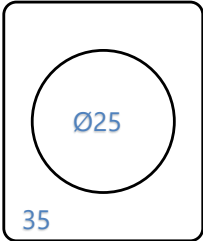
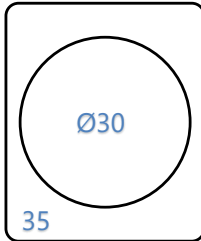
- Double-walled hopper also for filling side
- Piston lubrication system for one or both piston sides (drip lubrication)
- Different agitator systems for one or both hoppers
- Triple Shot (1 filling / 2 shells OR 2 fillings / 1 shell)
- Quatro Shot (2 fillings / 2 shells)
- Control of downstream machine (e.g. vibration table) possible via control panel
- Y 300 mm (standard 150)
- RFID for mould recognition



Variants with 9 or 12 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow shapes in widths of 261 ... 412 mm
 - For wide shapes in widths of 413 ... 517 mm
- Hopper Graduation Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to six different sizes (Ø6 ... Ø30 mm)

Piston Systems

Liner width 35 mm Piston stroke 52 mm				
Piston- Ø	6 mm 8 mm 10 mm	17 mm	25 mm	30 mm
Volume	... 4.1 ml ... 4.9 g	11.8 ml 14.2 g	25.5 ml 30.6 g	36.8 ml 44.1 g
Recommendations	Decorations or drops	Chocolates in One Shot, Triple Shot or Quatro Shot	Bars with or without small inclusions	Bars with or without small inclusions

Machine capacities (examples at 12 pistons per side, thus 24 pistons in total if as a boxer)

Example 1: 100g bars without filling

Maximum design with two chocolate hoppers (3 nozzles per bar)
 Per stroke 8 panels, 17 strokes/min and 60 min/h yields:

- **8,160 bars/h or 816 kg/h**

Example 2: 10g One Shot Chocolate

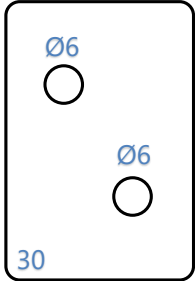
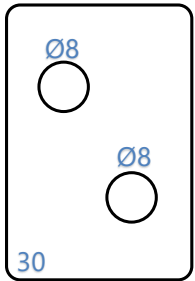
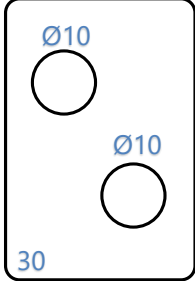
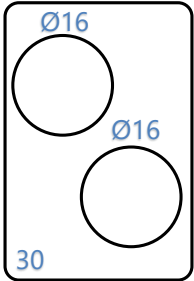
Maximum design with two hoppers (chocolate and filling)
 Per stroke 12 chocolates, 20 strokes/min and 60 min/h yields:

- **14,400 chocolates/h or 144 kg/h**

Variants with 36, 48 or 72 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow mould width from 261 ... 412 mm
 - For wide mould width from 413 ... 517 mm
- Hopper separation wall Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to four different sizes (Ø6 ... Ø16 mm)

Piston Systems

Liner width 30 mm Piston stroke 60 mm				
Piston- Ø	6 mm	8 mm	10 mm	16 mm
Volume	1.7 ml	3 ml	4.7 ml	12.1 ml
	2 g	3.6 g	5.7 g	14.5 g
Recommendations	Decorations very fine	Decorations	Decorations coarse	Chocolates in One Shot, Triple Shot or Quatro Shot

Machine Capacities (Examples with 72 pistons per side, 144 pistons in total, if as Boxer)

Example 1: 100g Bars without filling (solid)

- Possible but machine not ideal for bars

Example 2: 10g One Shot Chocolates

Maximum design with two hoppers (chocolate and filling)

Each stroke 72 chocolates, 20 strokes/min and 60 min/h results in:

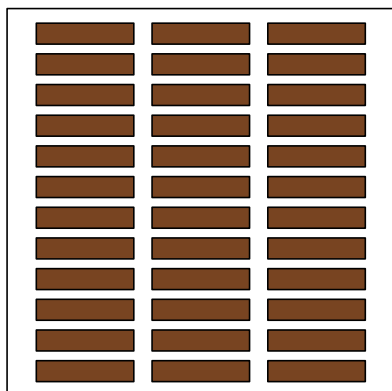
- **86'400 chocolates/h or 864 kg/h**



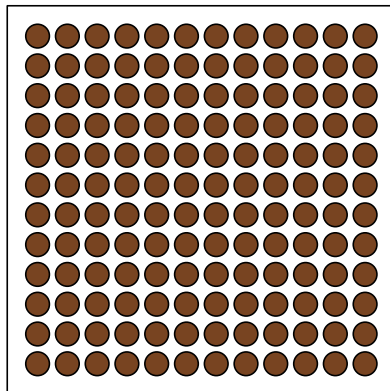
Combination options with 45 mm cylinder liner

With a liner width of 45 mm, the system is equipped with double pistons. Thus, for example, a single piston per liners can be equipped individually on the left side and double-piston liner on the right side. This increases variability. Here are examples with moulds in dimension 500 mm x 500 mm:

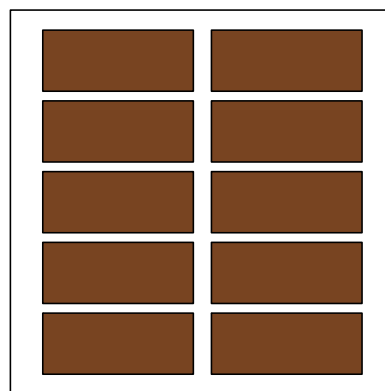
12x3 mould = 36 bars



12x12 mould = 144 Chocolates



5x2 mould = 10 blocks



For the alignment of the off-centred pistons, dummies can serve as placeholders

45 mm double piston

45 mm single piston

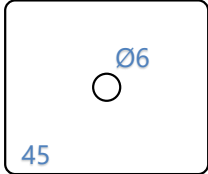
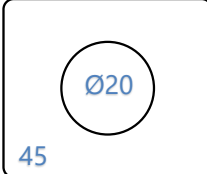
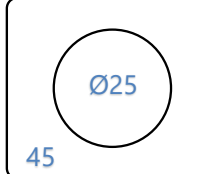
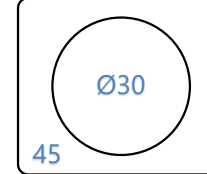


Piston ø mm	Stroke 52 mm	Piston volume		Stroke 60 mm			Piston volume	
		35	ml g	30/2	45/2	45	ml g	
6	X		1.47 1.76	X	X	X	1.70 2.04	
8	X		2.61 3.14	X	X	X	3.02 3.62	
10	X		4.08 4.90	X	X	X	4.71 5.65	
16				X			12.06 14.48	
17	X		11.80 14.16					
20					X	X	18.85 22.62	
25	X		25.53 30.63			X	29.45 35.34	
30	X		36.76 44.11			X	42.41 50.89	
40						X	75.40 90.48	

Variants with 9 or 12 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow mould width from 261 ... 412 mm
 - For wide mould width from 413 ... 517 mm
- Hopper separation wall Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to six different sizes (ø6 ... ø30 mm)

Piston Systems

Liner width 45 mm Piston stroke 60 mm				
Piston - Ø	6 mm 8 mm 10 mm	20 mm	25 mm	30 mm
Volume	... 4.7 ml	18.9 ml	29.5 ml	42.4 ml
	... 5.7 g	22.6 g	35.3 g	50.9 g
Recommendations	Lids or drops	Chocolates in One Shot, Triple Shot or Quatro Shot	Bars with or without small inclusions	Bars with or without small inclusions

Machine Capacities (Examples with 24 pistons each side, 48 pistons in total, if as Boxer)

Example 1: 100g Bars without filling (solid)

Maximum design with two chocolate hoppers (2 nozzles for one bar)

Each stroke 12 bars, 15 strokes/min and 60min/h results in:

- **10'800 bars/h or 1'080 kg/h**

Example 2: 10g One Shot Chocolates

Maximum design with two hoppers (chocolate and filling)

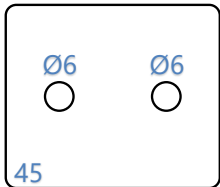
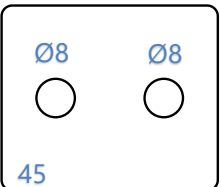
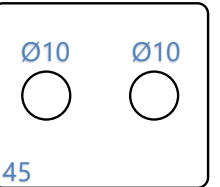
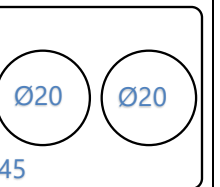
Each stroke 12 chocolates, 20 strokes/min and 60min/h results in:

- **14'400 chocolates/h or 144 kg/h**

Variants with 16, 18 or 24 pistons per side

- Axle variants
 - Z for standard fillings
 - X for more flexibility and control of the X-movement (belt running direction)
 - CAD for the maximum flexibility of all three axes. Ideal for decorations
- Operating variants
 - Single (for drops to bars without filling)
 - Boxer (for One Shot chocolates, bars or blocks)
- Range depending on mould widths
 - For narrow mould width from 261 ... 412 mm
 - For wide mould width from 413 ... 517 mm
- Hopper Graduation Chocolate Side and/or Filling Side
- Belt running direction from left to right or right to left
- Piston/liner pairs in up to four different sizes (Ø6 ... Ø20 mm)

Piston Systems

Liner width 45 mm Piston stroke 60 mm				
Piston - Ø	6 mm	8 mm	10 mm	20 mm
Volume	1.7 ml	3.0 ml	4.7 ml	18.9 ml
	2.0 g	3.6 g	5.7 g	22.6 g
Recommendations	Decorations very fine	Decorations	Decorations coarse	Chocolates in One Shot, Triple Shot or Quatro Shot

Machine capacities (examples at 24 pistons per side, thus 48 pistons in total if as a boxer)

Example 1: 100g bars without filling (solid)

- Possible but machine not ideal for bars

Example 2: 10g One Shot Chocolates

Maximum design with two hoppers (chocolate and filling)

Each stroke 24 chocolates, 20 strokes/min and 60 min/h yields:

- **28,800 chocolates/h or 288 kg/h**



Applications



- Bars without filling
- Bars with small inclusions up to 3 mm
- Bars with whole nuts



- Filled chocolates in 2 colours (One Shot)
- Filled candy bars in 2 colours (One Shot)
- Filled chocolates in 3 colours (Triple Shot)
- Filled chocolates in 4 colours (Quatro Shot)
- Filled chocolate bars (One Shot Curtain)



- Free forms without filling
- Free forms with fillings
- Free forms with two fillings



- Fillings for Macarons
- Fillings for Alfajores



- Decorations small 0.1g (eyes)
- Decorations big (body)

